

PRODUCTION-GRADE QUALITY ASSURANCE FRAMEWORK

Website & App Development Checklist

— Five phases from discovery to deployment —

Full-Stack Engineering

Next.js · React · TypeScript · FastAPI · PostgreSQL

Autonomous AI Workflows

LLM orchestration · RAG pipelines · MCP systems

Robotic Process Automation

Playwright headless bots at computational speed

Transparent Pricing

Published rates. Free working prototype before any payment commitment.

Engineering-First Culture

The founder writes production code personally on every client project.

Deterministic AI

Strictly typed scaffolding enforced around every probabilistic LLM call.

5 Phases

End-to-End QA

From discovery to go-live

30+

Checklist Items

Each mapped to production quality gates

Zero

Technical Debt

Every checkbox exists to prevent rework

COMPANY

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Phase 01

Strategic Discovery & Requirement Engineering

**Define the North-Star Business Objective**

Lead generation, SaaS subscription sales, internal data management, or brand authority — define the single north-star metric before any design begins. Everything downstream flows from this decision.

**Audience Persona Mapping**

Document technical proficiency, device usage (mobile vs desktop), and primary pain points of the end-user. Create at least three distinct personas with realistic use-case scenarios before the wireframing phase.

**Competitor UX Teardown**

Analyse 3 direct competitors. Note feature gaps, page speed scores (Lighthouse), and user journey friction points. These gaps become your conversion differentiators and must be documented before design begins.

**Brand & Aesthetic Guidelines**

Finalise colour hex codes (LIT Standard: Deep Navy #0D1B3E, Cyan #45D9D2), typography pairings (Space Grotesk + Inter), and SVG logo assets before any design tool is opened. Undefined brand identity causes costly rework.

**Data & Compliance Scoping**

Determine if the application will handle PII, requiring strict database encryption, DPDP Act 2023 compliance tracking, Row Level Security, and comprehensive audit-log architecture from day one of development.

PHASE 01 PRO TIP

Never skip the Competitor UX Teardown. Understanding where your competitors fail is more valuable than any feature list. Prioritise gaps in their user journeys — these become your highest-converting differentiators.

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Phase 02

UI/UX Architecture & Prototyping



Information Architecture & Sitemap

Map every required page, nested route, and API endpoint. Define the full navigation hierarchy before wireframing begins. Changes at this stage cost minutes; changes after development begins cost days.



Low-Fidelity Wireframing

Sketch core layouts for Dashboard, Landing Page, Authentication flow, and Settings — validated by the client before Figma work begins. Approval at this stage prevents expensive design rework later.



High-Fidelity Figma Prototyping

Design pixel-perfect mockups for Desktop and Mobile viewports. Enforce high contrast, WCAG 2.1 accessibility standards, and the LIT dark-mode aesthetic. Every interactive state must be designed before handoff.



PostgreSQL Database Schema Design

Map all tables (Users, Profiles, Transactions, etc.) and draw Entity-Relationship diagrams with all foreign key relationships and indexes. Getting the schema wrong at this stage costs 10x more to fix later.



API Contract Definition

Document the exact JSON structures passed between Next.js frontend and FastAPI backend. No ambiguity in the data contract. Both sides must agree on field names, types, and nullable properties before coding begins.

PHASE 02 PRO TIP

The Database Schema Design is the most critical deliverable of this phase. Architect for the data you will have in Year 2, not just Year 1. A poorly normalised schema generates irreversible technical debt.

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Phase 03

Core Engineering & Development



Environment Setup & Version Control

Initialise Next.js repository, configure Tailwind CSS, set up Git with branch strategy (main / develop / feature), and create the CI/CD pipeline before writing a single line of application code.



Authentication Implementation

Integrate Supabase Auth or NextAuth (OAuth, Email, Magic Links). Test all edge cases including password reset, token refresh, and session expiry before building any feature that depends on the user session.



Frontend Component Construction

Build modular, reusable React components (Buttons, Modals, Tables, Navbars) with full TypeScript typing. Components must be documented and tested in isolation before integration into page-level layouts.



Backend & API Endpoint Construction

Build secure FastAPI endpoints with Pydantic validation, rate limiting, and comprehensive error response schemas. Every endpoint must be tested against its API contract before the frontend consumes it.



Database Provisioning & RLS Policies

Deploy Supabase instance, run SQL table creation scripts, and enforce Row Level Security (RLS) policies on every table containing user data. Verify that JWT tokens correctly scope all data access.

PHASE 03 PRO TIP

Build authentication first, before any feature development. Every component downstream depends on the user session object. Authentication bugs contaminate all subsequent modules and are expensive to retrofit correctly.

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Phase 04

Quality Assurance, Security & Performance



Automated Unit & Integration Testing

Write unit tests for all critical business logic — pricing calculators, data parsers, authentication flows. Target 80%+ code coverage on core functions before the QA phase begins.



Security Penetration Testing

Attempt SQL injection, test RLS policies for cross-tenant data leaks, verify JWT token integrity, and audit all public API endpoints. Every security finding must be remediated before production deployment.



Cross-Browser & Device Testing

Verify flawless rendering on Chrome, Safari, Firefox, iOS, and Android. Test on actual physical devices, not only browser DevTools simulations, for accurate touch interaction and viewport behaviour.



Lighthouse Performance Optimisation

Achieve 90+ scores in Performance, Accessibility, and SEO. Optimise image formats to WebP, implement lazy loading, enable SSR caching headers, and eliminate render-blocking resources before production deployment.



Form Validation & Edge-Case Testing

Test every user input form for correct error handling and safe database writing. Deliberately test edge cases including empty inputs, special characters, XSS payloads, and concurrent write conflicts.

PHASE 04 PRO TIP

Test with real user data samples (anonymised). Edge cases only appear with real-world entropy. Testers using predictable inputs will miss the bugs that production users find within the first 24 hours of go-live.

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